

The Topology of Aliasing

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Abstract

Apparent multiplicity emerges from temporal undersampling of continuous processes. When resolution increases, discrete points collapse to continuous structure. When resolution decreases, continuous structure is reconstructed as discrete points. Observed structure appearance depends on sampling rate relative to process rate.

We respect observations.

We do not inherit interpretations.

1. Measurement Constraints

1.1 Discrete Sampling

All measurement samples discretely. Continuous processes are captured at discrete intervals. No measurement apparatus records continuously. Every instrument produces discrete outputs at finite temporal resolution.

1.2 The Resolution Relationship

Sampling rate relative to process rate determines appearance. The same process produces different observations at different sampling densities. When sampling rate falls below process rate, information is lost. Shannon-Nyquist [1,2] establishes this as a constraint, not interpretation: faithful reconstruction requires sampling at twice the highest frequency component. Below this threshold, aliasing is inevitable.

1.3 Scope

What follows is a logical analysis of resolution-dependent structure. The reasoning proceeds from measurement theory alone. Observations are separated from interpretations throughout.

2. When Resolution Increases, Continuity Replaces Multiplicity

2.1 Single Frame Observation

One discrete measurement produces one position. No single measurement shows multiple simultaneous positions. This is what measurement produces: a discrete outcome at a discrete moment.

2.2 Aggregation Across Measurements

Multiple discrete measurements produce multiple positions. Extended structure requires temporal aggregation. What is reconstructed as multiple entities derives from sample aggregation, not primary observation. The aggregation occurs after measurement, not during it.

2.3 The Symmetry

Undersample a continuous process: discrete points appear. Sample sufficiently: continuous structure appears. Same process, different measurement cadence. The transition from apparent multiplicity to apparent continuity follows directly from increasing sampling density. No threshold separates these regimes. They form a continuous spectrum.

3. Resolution-Dependent Appearance

3.1 High Sampling Density

At high sampling density, continuous structure becomes visible. Extended regions, gradients, and smooth distributions appear. Intermediate states are captured. The appearance is smooth because the sampling rate exceeds the rate at which the process evolves between distinguishable states.

3.2 Low Sampling Density

At low sampling density, discrete points become visible. Separated loci and apparent multiplicity appear. Intermediate states are not captured. The appearance is discontinuous because the sampling rate falls below the rate at which the process evolves between distinguishable states.

3.3 The Transition

These are not two different entities. One continuous process produces different appearances at different sampling rates. The progression from single-point observations to multiple points to extended structure follows from increasing resolution. At each stage, what changes is sampling density, not the process being sampled.

4. Logical Consequences

4.1 Apparent Multiplicity

A continuous process sampled at discrete moments produces discrete outcomes. Each sample yields one position. Aggregation across samples creates the appearance of multiple positions. No simultaneous occupation is required to produce this appearance. The multiplicity is in the aggregated record, not in any single measurement.

4.2 Extended Structure

Extended structure emerges when sampling density is sufficient to capture intermediate states. It disappears when sampling density is insufficient. The appearance of extension is resolution-dependent. What is observed as extended at one sampling rate is observed as discrete at another.

4.3 Sampling Artifacts

Discontinuity arises from undersampling. Apparent stasis arises from oversampling. Both are measurement effects, not process properties. The artifact type depends on the direction of sampling-process rate mismatch.

5. Minimum Sufficient Description

5.1 What Measurements Show

Measurements show discrete outcomes. Aggregation across measurements shows patterns. Phase relationships between samples produce interference patterns. No measurement shows simultaneous occupation of multiple positions in a single frame. Every appearance of multiplicity derives from aggregation.

5.2 Sufficient Explanation

A continuous process exists. We sample it discretely. Appearance depends on sampling rate relative to process rate. Nothing additional is required. This description is complete with respect to what measurements produce.

5.3 What This Removes

No ontological distinction between discrete entities and continuous fields is required. No special status for measurement-dependent appearance is required. No dual descriptions of the same phenomenon are required. No additional interpretive layers are required. The measurement constraints are sufficient.

6. Measurement Extremes

6.1 Undersampling

When sampling rate falls below process rate, continuous evolution is observed as discontinuous. Only endpoints are captured. Intermediate states remain unresolved. Aliasing artifacts are inevitable. The appearance of discontinuity is proportional to the degree of undersampling.

6.2 Oversampling

When sampling rate exceeds process rate, evolution is observed as static. The same state is captured repeatedly. Changes occur below detection threshold. Saturation artifacts appear. The appearance of stasis is proportional to the degree of oversampling.

6.3 Symmetry

Same mechanism. Opposite sampling direction. No boundary separates these regimes. The spectrum of resolution is continuous. Undersampling and oversampling are positions on a single axis, not distinct phenomena.

Effect	Undersampling	Oversampling
Sampling relation	Below process rate	Above process rate
Appearance	Discontinuous	Static
Unresolved structure	Between samples	Below threshold
Observable	Endpoints only	Persistence only
Dominant artifact	Aliasing	Saturation
Underlying evolution	Continuous	Continuous

7. Conclusion

When resolution increases, multiplicity collapses into continuity. When resolution decreases, continuity is reconstructed as multiplicity. Structure appearance depends on sampling rate relative to process rate.

The constraint chain is complete:

1. Measurement samples discretely.
2. Continuous processes exist.
3. Shannon-Nyquist applies.
4. Appearance depends on sampling rate relative to process rate.
5. Structure appearance is resolution-dependent.

No additional ontological categories are required. The measurement constraints are sufficient.

References

- [1] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948.
- [2] Harry Nyquist. Certain topics in telegraph transmission theory. *Transactions of the American Institute of Electrical Engineers*, 47(2):617–644, 1928.